MP352
Special Relativity

Time allowed: $1\frac{1}{2}$ hours

Answer **ALL** questions

This is a **SAMPLE** exam, roughly reflecting the general structure of the finals for 2016 – 2017. Remember that **ALL** questions are to be answered.

1. Consider 4×4 matrices Λ which satisfy the following relation

$$\Lambda^T g \Lambda = g \,, \tag{1}$$

where g is the metric tensor.

(a) Write down the metric tensor g in the the two common conventions.

[4 marks]

(b) Write down the 4×4 matrix transforming spacetime coordinates (ct, x, y, z) of an event under a Lorentz boost in the x direction. Show that this matrix satisfies condition (1).

[15 marks]

(c) Show that, if a transformation of spacetime coordinates (ct, x, y, z) preserves the Minkowski norm, then it must satisfiy condition (1).

[16 marks]

(d) Show that the set of matrices which satisfiy condition (1) forms a group under matrix multiplication. Verify all four group properties.

[20 marks]

2. (a) A rocket moves at speed u directly away from earth. It ejects a shuttle at speed v perpendicular to the direction of its motion. Measured with respect to earth, what are the velocity components of the shuttle? Measured with respect to earth, what is the speed of the shuttle?

[13 marks]

(b) The kinetic energy of motion of a particle is the relativistic total energy minus the rest energy. A particle has rest mass M and speed v. If $v \ll c$, then show that the kinetic energy due to motion is approximated by the well-known non-relativistic expression for the kinetic energy.

Derive the leading correction to the non-relativistic expression.

[12 marks]

(c) A photon with energy E collides with a stationary mass m. They combine to form one particle. What is the mass of this particle? What is its speed?

[20 marks]